

Milestone Four Narrative

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CS 499 Computer Science Capstone

The artifact selected for this milestone is the Python-based CRUD module created for the CS 340 Animal Shelter project. This module provides a structured interface for creating, reading, updating, and deleting documents in the AAC MongoDB collection. The artifact was originally developed during CS 340 and has been carried forward into CS 499 as a consistent ePortfolio artifact that can be enhanced across categories. For Milestone Four, the enhancement focuses on the databases category, emphasizing data access efficiency, database performance considerations, and reliable interaction patterns when working with a persistent data store.

This artifact is included in my ePortfolio because it demonstrates practical, real-world database interaction within a software system. While the original module provided functional CRUD access, the database-focused enhancement improves how queries are executed and how results are returned. Specifically, the artifact was improved by adding support for query shaping features that directly affect database performance and usability, including projection, optional sorting, and optional limiting of results. These changes reduce unnecessary data transfer and help prevent accidental full collection reads during interactive analysis or dashboard usage. In addition, index creation support was added to improve performance for common filter fields used in dashboard-style queries. Together, these improvements demonstrate database competency by showing how query patterns, indexing, and result shaping contribute to efficient and reliable systems.

This enhancement meets the course outcomes I planned to address in Module One for the databases category by demonstrating intentional design decisions related to database access,

performance, and data handling. My outcome-coverage plans do not require changes at this time because the enhancement remains consistent with the approved artifact and the progression established in earlier milestones.

Enhancing this artifact reinforced that database competency is not limited to connecting to a database and running queries but also includes designing query behavior that is efficient and predictable. One challenge was ensuring that performance-focused improvements did not alter the module's expected behavior or introduce scope drift beyond the artifact's purpose. This process strengthened my understanding of the practical relationship between query design, indexing, and application-level reliability. It also highlighted the value of implementing database improvements in a controlled and well-documented way so that the artifact remains maintainable and clearly communicates design intent to a technical reviewer.